Zatomic

ZECATHON · CROSS-CHAIN · P2P OTC

Swap shielded ZEC for ETH.

No bridge. No wrapped ZEC.

An atomic swap for two parties who have already agreed a trade.

162-55-186-88.sslip.io · github.com/yugocabrio/zatomic

THE PROBLEM

Why a shielded atomic swap is hard

01

Neither side wants to go first

Whoever pays first can be cheated, so both payments must happen, or neither.

02

Shielded notes have no scripts

No hash locks or time locks: a shielded note moves only with a valid signature from its key.

03

Other routes add trust or exposure

Bridges must prove Zcash history to Ethereum; others rely on solvers, public viewing keys or transparent ZEC.

> Zatomic keeps the ZEC shielded and asks Ethereum to verify **no Zcash history**.

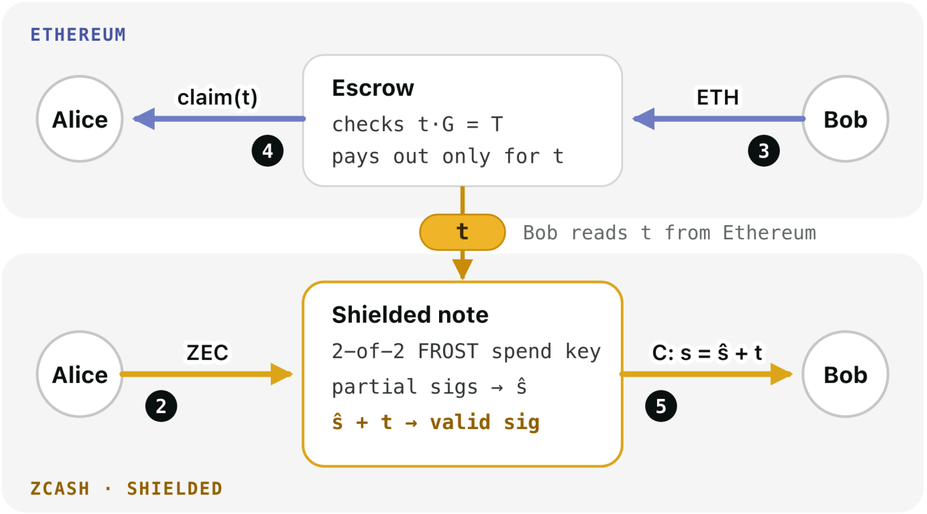
HOW IT WORKS

Two payments, one secret

1 **Off chain, before funds move: both sign with the 2-of-2 FROST key**

C, paying the ZEC to Bob: their joint signature ŝ needs t to be valid

R, returning it to Alice: fully signed, but not valid until a deadline



IF NOBODY REVEALS t

After the deadlines: Bob takes his ETH back, Alice's refund returns her ZEC.

1 Off chain, before any funds move: Bob's claim needs t, Alice's refund needs a deadline.

2 Alice locks ZEC in a note whose spend key is a 2-of-2 FROST key.

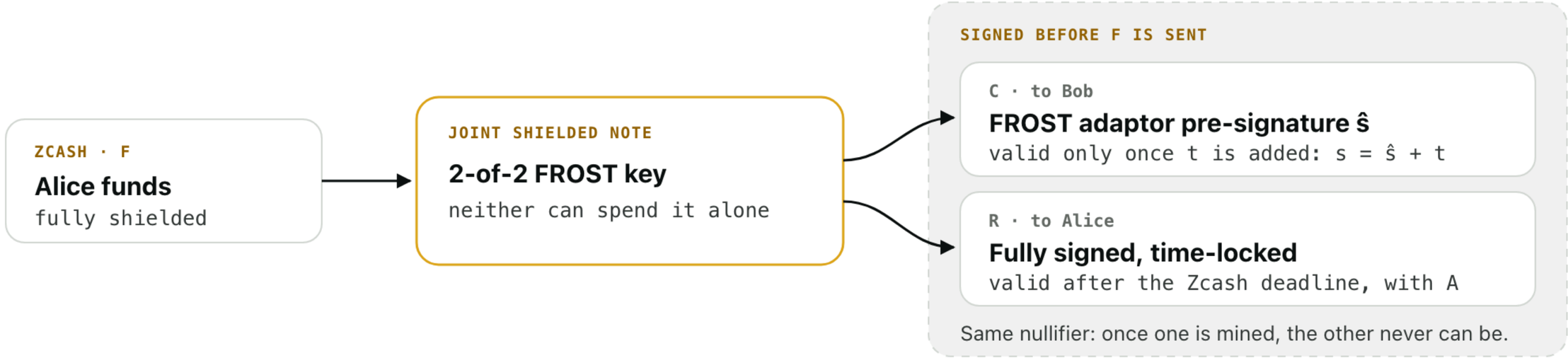
3 Bob locks ETH in the escrow, bound to T = t·G.

4 Alice takes the ETH by revealing t.

> **5** The same t completes their FROST adaptor pre-signature: **s = ŝ + t**.

PREPARED BEFORE FUNDING

One note, two exits, both signed first



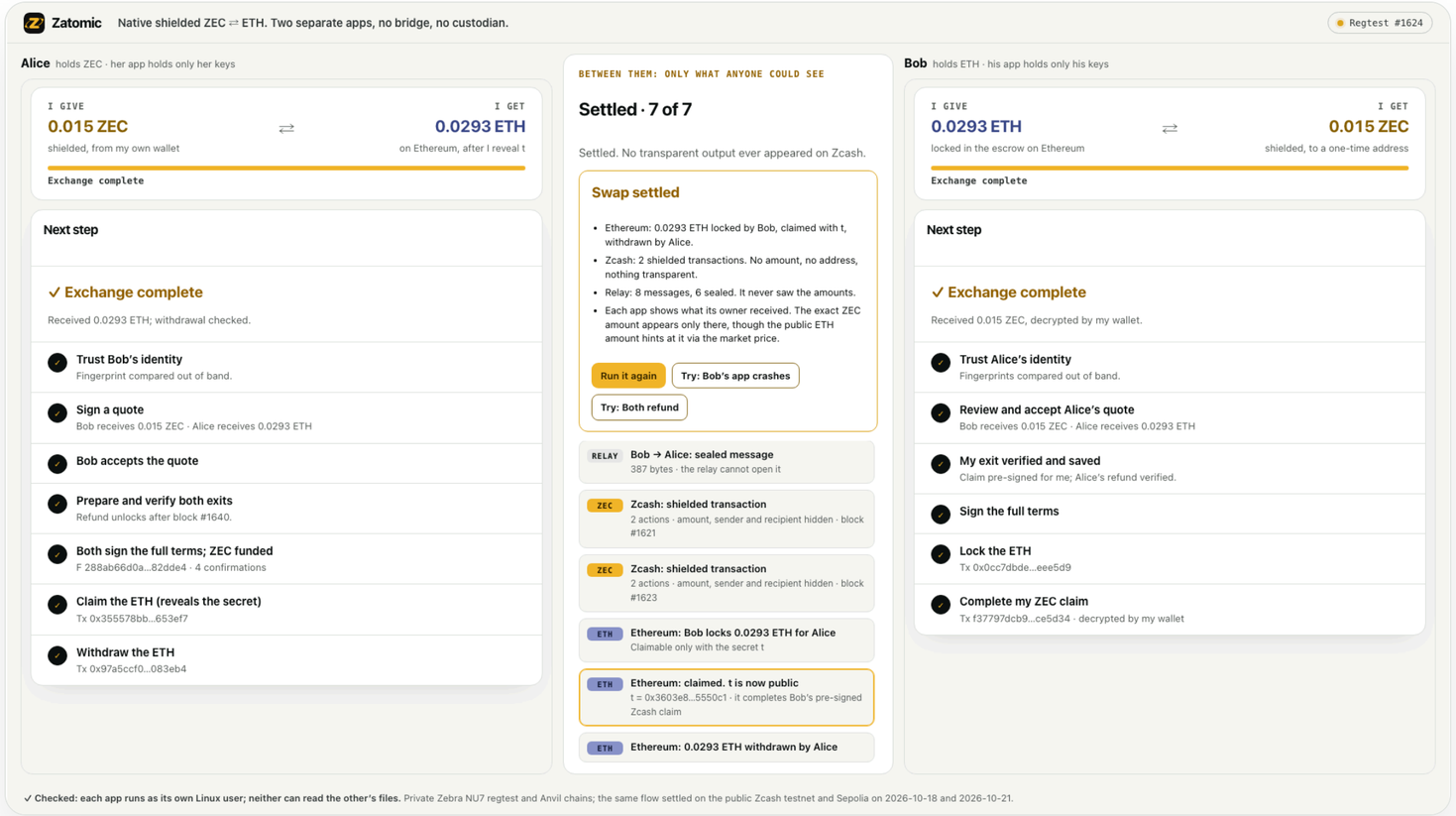
Signed before the note exists: in Zcash v6, signatures do not cover the anchor or the proof.

A settled swap stays fully shielded. Only a refund uses one small transparent coin from Alice’s own change.

LIVE DEMO

Two separate apps. Only public data between them.

Real Zebra NU7 and Anvil transactions on private test chains. Each app holds only its owner’s keys.



PUBLIC TESTNETS · ZCASH TESTNET + SEPOLIA

Both outcomes, on real networks

4

funded public runs: two settled, two refunded both sides

2

Zcash transactions in a settled swap, both fully shielded

−79%

Ethereum gas per settled swap with the shipped V2 escrow

Every transaction is linked in the records: github.com/yugocabrio/zatomic

WHAT IS NEW

Zcash’s own spend authorization, as one side of an atomic swap

- A RedPallas adaptor signature on a 2-of-2 FROST key
- Ethereum checks t on Pallas directly: no bridge, no cross-curve proof
- If the trade fails, Alice’s refund needs nothing from Bob: Zcash enforces its lock time

`Live demo · 162-55-186-88.sslip.io` `Code · github.com/yugocabrio/zatomic`